

Split-Complex Numbers as Imaginary-Radius Orthogonal Euler Circles (Exercise XXVII)

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Abstract

We report the formal implementation of an observation frame suggested by the pat exercise series: two orthogonal Euler circles, one of radius 1 and one of radius i — an imaginary-radius circle, which cannot exist in the complex plane (radii are nonnegative reals, $|z| = i$ has no solution). The host structure is the split-complex numbers $a + bj$, $j^2 = +1$, with pseudo-norm $\|a + bj\|^2 = a^2 - b^2$ (Minkowski metric): the radius-1 circle is the hyperbola $a^2 - b^2 = 1$, and the radius- i circle is the conjugate hyperbola $a^2 - b^2 = -1$, nonempty (its unit j lies on it). Seven theorems are machine-checked in Lean 4 / mathlib (with a self-built 30-line split-complex structure, standard mathematics): $j^2 = 1$, pseudo-norm multiplicativity $\|zw\|^2 = \|z\|^2\|w\|^2$, nonemptiness of both circles, the orthogonality condition $1^2 + i^2 = 0$, disjointness of the conjugate hyperbolas, and the observation contrast: every odd prime is a split pseudo-norm $p = ((p+1)/2)^2 - ((p-1)/2)^2$, whereas in the complex plane only primes $p \equiv 1, 2 \pmod{4}$ are norms (Fermat). Zero errors, zero sorry.

Keywords: Lean, formalization, split-complex numbers, hyperbola, imaginary radius, 0pat

1 The observation frame

The exercise series has been observing the prime axis through circles: the circle $\mathbb{Z}/p\mathbb{Z}$ (Exercise XXIV), the orthogonal coordinate frame of two circles (Exercise XXV), and the critical circle of the Riemann direction. The question posed here is whether the two orthogonal Euler circles may carry *different units* — one of radius 1 (natural unit) and one of radius i (imaginary unit) — without forcing the Cartesian constraint of the complex plane, where the real and imaginary axes share the same unit.

The answer is that such a frame cannot live in the complex plane: a circle $\{z : |z - c| = r\}$ requires $r \geq 0$, and $|z| = i$ has no solution. The host structure is the *split-complex* numbers, the two-dimensional real algebra $a + bj$ with $j^2 = +1$ (contrast $\mathbb{C}$, where $J^2 = -1$). Its pseudo-norm is the Minkowski metric $\|a + bj\|^2 = a^2 - b^2$. In this structure, the radius-1 circle is the hyperbola $a^2 - b^2 = 1$, and the radius- i circle is the conjugate hyperbola $a^2 - b^2 = -1$, which is nonempty — its unit j lies on it. The orthogonality of the two radii is exact in the split metric: $r_1^2 + r_2^2 = 1^2 + i^2 = 1 + (-1) = 0$, the condition for orthogonal concentric circles.

2 The theorems

Seven theorems, all machine-checked with zero sorry. The split-complex structure is defined in 30 lines (a pair of reals with multiplication $(a, b)(c, d) = (ac + bd, ad + bc)$); the theorems are standard mathematics.

The defining algebra (*j_sq*). $j^2 = 1$. This is the single structural difference from the complex numbers ($J^2 = -1$) and the origin of every subsequent property.

Pseudo-norm multiplicativity (*normSq_mul*). $\|zw\|^2 = \|z\|^2\|w\|^2$. The split pseudo-norm behaves like the complex norm under multiplication; the proof is a direct expansion.

Nonemptiness of both circles. $\|j\|^2 = -1$ (the radius- i circle contains its unit j) and $\|1\|^2 = 1$ (the radius-1 circle contains 1). The imaginary-radius circle exists and is inhabited — the property that fails in $\mathbb{C}$.

Orthogonality (*orthogonal_radII*). $1 + (-1) = 0$: in the split metric, the radii 1 and i satisfy $r_1^2 + r_2^2 = 0$, the concentric-circle orthogonality condition. The two Euler circles are orthogonal in exactly the sense the frame requires.

Disjointness (*circles_disjoint*). No point lies on both circles: $a^2 - b^2 = 1$ and $a^2 - b^2 = -1$ cannot hold together. The conjugate hyperbolas are disjoint — the frame has two separate loci, which is what makes them a coordinate pair rather than one circle.

Observation contrast (*odd_prime_is_split_norm*). Every odd prime p is a split pseudo-norm: $p = ((p+1)/2)^2 - ((p-1)/2)^2$. In the complex plane, only primes $p \equiv 1, 2 \pmod{4}$ are norms (Fermat's two-square theorem, Exercise XXIV). The split frame therefore observes primes without discrimination — every odd prime is representable — whereas the complex frame distinguishes half of them. The contrast is the frame's first observational result: the choice of unit (1 vs i) determines the discriminating power of the observation.

3 Honest boundary and position

As in every exercise: the content is classical mathematics (the split-complex numbers are a standard structure; all theorems are known), the proofs are machine-checked, and no claim is made beyond the statements. The value is positional: the frame requested by the series (radius-1 and radius- i orthogonal Euler circles) is now implemented and its basic

observational property (no prime discrimination) is established. The frame is available for further observation — for example, of structures that the complex plane cannot host.

4 Formalization

The proof is a single Lean file (proof.lean, 135 lines), importing only Mathlib.Data.Real.Basic, Mathlib.Data.Nat.Prime.Basic, Mathlib.Tactic.Ring, and Mathlib.Tactic.Linarith. The split-complex structure (SplitComplex, 30 lines) is a self-contained standard definition; no pat framework concepts are used. Compiled with Lean 4.32.2 / mathlib v4.32.2, zero errors, zero sorry.

5 Conclusion

The imaginary-radius orthogonal Euler frame is implemented: split-complex numbers host the radius- i circle, the orthogonality condition $1^2 + i^2 = 0$ holds exactly, and the frame's first observation is that it does not discriminate primes (every odd prime is a pseudo-norm), in contrast to the complex plane. The frame now exists, machine-checked, ready for further observation.

References

Theorem	Statement	Proof core
j_sq	$j^2 = 1$	ext; simp
normSq_mul	$\ zw\ ^2 = \ z\ ^2 \ w\ ^2$	simp; ring
imaginary_radius_circle	$\ j\ ^2 = -1$	simp
real_radius_circle	$\ 1\ ^2 = 1$	simp
orthogonal_radII	$1 + (-1) = 0$	norm_num
circles_disjoint	$\neg \exists z, \ z\ ^2 = 1 \wedge \ z\ ^2 = -1$	linarith
odd_prime_is_split_norm	$p = ((p+1)/2)^2 - ((p-1)/2)^2$	div_add_mod; omega